

MESO-SCALE STANDARD EVAPOTRANSPIRATION ‘CLIMATE’ CLASSIFICATION DERIVED FROM NUMERICAL WEATHER PREDICTION MODELS AND ARTIFICIAL INTELLIGENCE

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ABSTRACT

The Weather, Research and Forecasting (WRF) model and Self-Organizing Maps (SOM) artificial neural network (ANN) have been used to classify regional evapotranspiration (ETo) ‘weather’. Here, this concept is expanded to develop a pilot ETo ‘climate’ classification system in a regional subset of the Andes and the Amazon. ETo ‘climate’ is defined as the frequency of ETo ‘weather’ classes passing through a geographical location (*i.e.*, a pixel in the WRF model), allowing the construction of a histogram for each pixel. The histogram is then used as a one-dimensional signal for another SOM classification, offering a regional perspective on how ETo behaves and specifically what a geographic location can expect in terms of ETo variables, similar to the Köppen climate classification system. This research presents the results of several classification comparisons to produce a repeatable pilot system for Eto ‘climate’ classification and a map that offers the potential to improve irrigation decision-making.

Index Terms— Evapotranspiration, Self-Organizing Maps, neural network, Weather Research and Forecast model, ETo classification

1. INTRODUCTION

Global population is projected to reach 8.5 billion by 2030 [1], primarily concentrated in developing countries. Food production may not meet the growing demands given by anticipated climate change scenarios. Furthermore, rising temperatures are predicted to alter precipitation patterns, leading to droughts, floods, and the melting of glaciers and permanent snow-capped mountains, which are the primary freshwater source for example for the Andean region [2].

Irrigation enables food production in dry and semi-arid regions; however, its unsustainable use can accentuate water scarcity, and can cause soil chemical imbalance, contamination of water sources, and loss of topsoil [3].

Efficient irrigation practices, considering evapotranspiration (ETo) dynamics, can minimize these issues. The Penman-Monteith equation [4] calculates ETo, and helps to estimate the crop water requirements. By understanding the crop’s water needs and by providing accurate irrigation, many of the negative impacts can be reduced.

Estimation of ETo using the Penman-Monteith equation faces challenges in developing countries due to limited and unreliable local meteorological data [5]. However, these challenges can be overcome by integrating remotely sensed information and numerical weather prediction models, which provide spatially distributed proxy data [6]. Additionally, data-driven models using artificial neural networks (ANN) have proven successful in estimating ETo [7], [8].

The WRF model from the University Corporation for Atmospheric Research (UCAR) and the National Center for Atmospheric Research (NCAR) atmospheric laboratories in the United States [9] can provide proxy variable values required to estimate the Penman-Monteith equation. WRF uses Global Forecast System (GFS) [10] in six-hour intervals and, simply put, interpolates the data to a user-specified interval. Every six hours, GFS releases a new set of forecast data that predict what is likely to happen with the weather, extending hundreds of hours into the future. GFS also simultaneously reanalyzes the previously released start data using satellites, weather stations, and ocean buoys for a better rendering of the weather events in that time frame. When numerical prediction models use reanalyzed data this way, it is called hindcasting rather than forecasting.

The aim of this research is to achieve a repeatable ETo ‘climate’ classification system using five years of WRF and GFS reanalyzed data. The methodology and the resulting map contribute to the understanding of ETo dynamics that can be used for the efficient management of water.

2. METHODOLOGY

2.1. Data Sources

Data for ETo ‘weather’ classification were locally generated in Ecuador using the Imbabura High-Performance Computing (HPC) complex at Yachay Tech University using Linux, Python 3, MiniSom package [11], NumPy [12], and the Python On-Line Imaging (POLI) package [13] among other utilities. For the ETo ‘weather’ classification, 0.25-degree GFS reanalyzed data were used to generate hourly hindcast data, over five years (2017 – 2022), for a subset of the Andean and Amazonian regions using 171 x 171 pixels at a 3.3 km resolution, as in [14].

2.2. ETo ‘climate’ classification

For every geographic pixel, a one-dimensional and normalized signal was constructed using the variables (and their hourly values) needed to calculate ETo using the hourly Penman-Monteith equation. This feature space was presented to SOM for unsupervised classification.

After defining Eto ‘weather’ classes, the dynamics of these classes over time were analyzed [15]. The count of classes that a geographic pixel experiences was identified and analyzed through the construction of histograms.

Histograms were presented to SOM again, with different hyperparameters, to discriminate or cluster ETo ‘climates’. Every geographical pixel has a corresponding histogram of weather classes (25 in this case), that defines the feature space for SOM to cluster.

To construct this feature space, the previously derived ‘weather’ class weights were applied to hourly data over five years for each pixel, giving integer ‘weather’ class labels for every day, stored in a 171 x 171 x 1,826 buffer. For every pixel, a histogram of class labels is rendered, giving a 171 x 171 x 25 data buffer. This buffer is then presented to SOM for ‘climate’ clustering and is not normalized since all histogram bins (class labels) have the same units (number of occurrences) as in Figure 1.

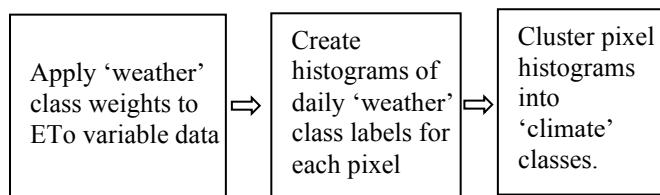


Figure 1. Flowchart for Eto ‘climate’ classification.

2.3. ‘Weather’ Class Repeatability

‘Climate’ classification repeatability depends on the accuracy of ‘weather’ classification. As SOM is based on random sampling (a stochastic process), a modified Cramer’s-V similarity measure [16] was applied to compare different ‘weather’ classification runs. As a result, one-to-one mapping between run classes were created and used to implement ‘climate’ repeatability.

3. RESULTS

The concept of ETo ‘climate’ has been developed by analyzing histograms built upon frequencies of ETo ‘weather’ classes passing through a geographic pixel. As expected, only a certain set of classes pass through a pixel over time.

The model hyperparameters selection was derived through trial and error to achieve repeatable values. Accordingly, a 3 x 4 neural grid (12 ETo ‘climate’ classes), a learning rate of 0.3, and 30 epochs were selected. More details on parameters and usage is available through source code at [17], while output data can be found at <https://zenodo.org/deposit/7972038>.

In addition, repeatable Cramer’s-V similarity values for the ‘weather’ classification, were on average 0.97 (ranging from 0 to 1) with unity being an exact match [18]. There existed a concern with repeatable ‘climate’ runs, but this time a new layer had to be considered.

For example, two ‘weather’ runs, W_1 and W_2 , with a Cramer’s-s-V similarity of 0.98 will yield slightly different outputs when the W_1 and W_2 class weights are applied to the same data. If ‘climate’ SOM runs using W_1 and W_2 outputs are applied, rigorous comparison of the ‘climate’ output, C_1 and C_2 , is not possible using Cramer’s-s-V value due to W_1 and W_2 outputs being different. To use Cramer’s-s-V measure, either W_1 or W_2 outputs should be chosen and then an analysis of how the C_1 and C_2 class weights measure in similarity should be performed.

Instead, the one-to-one mapping of ‘weather’ classes given by the Cramer’s-V implementation was used to make W_1 and W_2 have the same class label regime, and then a ‘climate’ clustering was performed on their outputs, giving a more solid comparison approach. Here, C_1 was subtracted from C_2 and the number of pixels that were different was used as a percentage that indicated an inverse measure of similarity. Still, due to the random assignments of ‘climate’ class labels a class mapping was necessary before calculating the difference.

Figure 2 illustrates two ‘climate’ runs and their differences.

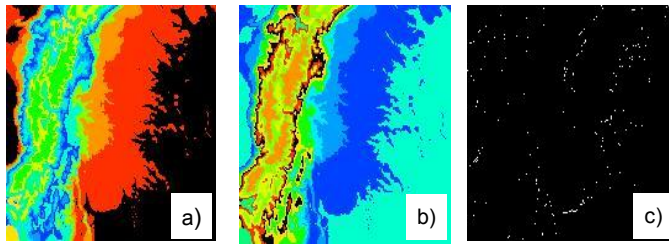


Figure 2. Obtained 'climate' maps and their difference.

- a) 'Climate' output C_1 derived from 'Weather' run W_1 ;
b) 'Climate' output C_2 derived from 'Weather' run W_2 ;
c) $C_2 - C_1$ resulting in 0.63% pixel difference.

For the 'climate' comparison, two sets of nine 'weather' runs each were used, with one-to-one Cramer's-V similarity values of 0.99. Each run employed five years of hourly data. For each 'climate' set, 36 one-to-one comparisons were made.

The results of the 36 one-to-one classification comparisons are presented in Table 1; and Figures 3 and 4 show the percentage of pixel difference distribution (Y-axis) for each one-to-one classification comparison (X-axis).

Table 1. Pixel difference as percentage between comparisons of 'climate' classification runs for the first and second sets.

Comparison	1	2	3	4	5	6	7	8	9
First	0.47	1.07	0.40	0.76	0.48	0.51	0.37	0.83	0.41
Second	0.39	1.50	0.65	1.13	0.47	0.62	0.60	0.54	1.32
Comparison	10	11	12	13	14	15	16	17	18
First	0.57	0.87	0.88	0.45	0.59	0.63	2.95	0.90	0.39
Second	0.35	1.56	0.57	0.84	0.49	0.60	0.61	0.79	0.42
Comparison	19	20	21	22	23	24	25	26	27
First	0.87	0.45	1.36	1.39	0.71	0.57	0.41	0.60	0.30
Second	0.60	0.96	2.52	0.90	0.55	0.65	0.66	0.40	0.73
Comparison	28	29	30	31	32	33	34	35	36
First	0.76	0.71	1.17	0.66	0.66	0.94	0.71	1.10	0.70
Second	0.49	0.67	1.50	0.32	3.54	0.57	1.75	0.82	0.77

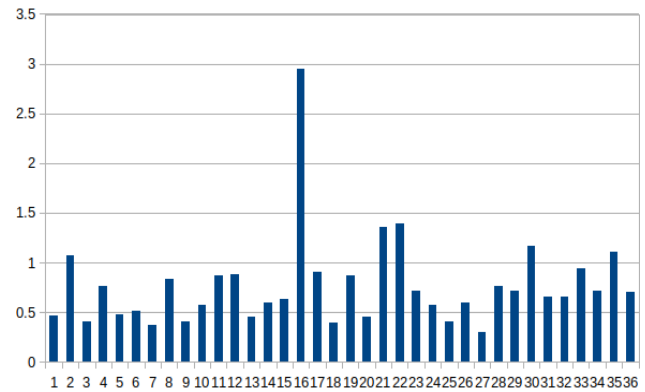


Figure 3. Pixel differences for comparisons of 'climate' classification runs for the first set, as per Table 1 values.

Min = 0.30%; Max = 2.95%; Avg = 0.77%

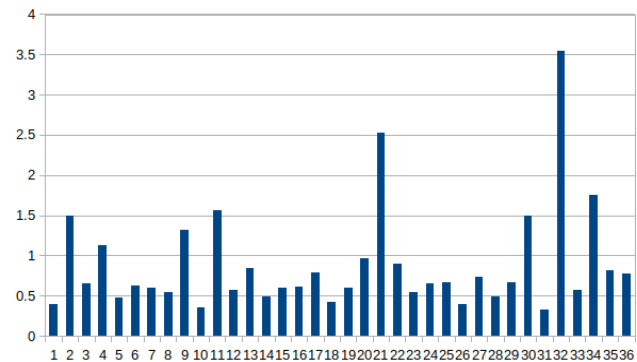


Figure 4. Pixel differences for comparisons of 'climate' classification runs for the second set, as per Table 1 values.

Min = 0.32%; Max = 3.54%; Avg = 0.88%

4. CONCLUSIONS

A method for characterizing the behavior of ETo has been presented in both a 'weather' and 'climate' context. It is important to emphasize that the employed methods in this study are based on stochastic principles, which inherently introduce some degree of error that should be recognized as inevitable. Rigorous efforts have been taken to minimize this error, resulting in an average pixel difference of less than one percent in assigning climate classes across multiple runs.

This research presents the results of several classification runs. Repeatable model hyperparameters were selected to obtain an ETo 'climate map'. This output emphasizes the significance of variable value ranges and their behavior, as better indicators of environmental dynamics, which are not fully captured when directly utilizing the Penman-Monteith equation.

Additional studies are necessary to validate these findings through ground-truthing and in conjunction with the adjustments of variables, in particular the number of 'weather' and 'climate' classes used. Also, the ETo 'weather' class

histogram does not contain information as to when the ETo ‘weather’ class event occurred and, therefore, loses the temporal component, which will be addressed in future studies.

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